

linux kernel monkey log

Linux Driver Project Status Report as of June 2009

This is a status report for the Linux Driver Project as of June 2009, describing what has happened in the past year of work. It was originally posted on the Linux Driver Project developer mailing list.

The Linux Driver Project (LDP) is now over two years old. It sprang up out of the complaints from some users and companies that there was a real "Linux driver problem". The perception was that Linux did not have good driver support, and that closed source drivers were potentially taking over some device types.

It turns out, that almost all hardware is currently already supported on Linux, with almost all major vendors shipping Linux drivers.

So last year, I proposed the following goals:

1. Continue to write new drivers for any company that wishes to have them developed. These drivers are all to be released under the GPLv2 and included in the main Linux kernel source tree hosted at kernel.org. Future maintenance of these drivers will be done either by members of the original company, or by the community members, depending on the wishes of the company.
2. Continue to be a focal point for companies to learn about the Linux development process and become part of the kernel community if they wish to. I hope to enlist more people to help out with this process, but if not, my airline miles logged will continue to increase.
3. Work more in an open development manner, hosting all experimental and development code in a public git tree, getting daily testing in the linux-next tree on all architectures.

Goal one (write drivers) has been very successful. Myself and many other members of the LDP have written new drivers for a wide range of different hardware devices, and gotten them merged into the main kernel tree. Several more are currently under development and we are averaging about 2 queries a month for different drivers from different companies.

This work will continue to happen in the following year, as everyone involved seems to be happy with it. However, there will be a few procedural changes in how this is working, to help resolve some of the issues that have occurred. For more details about this, see the discussion on the LDP development mailing list.

Goal two (educate people) has been reasonably successful. While working with many companies, we have helped them become maintainers of their own codebases within the kernel, integrating them into the main kernel development process, allowing them to control their drivers and direction for their Linux support much easier. However the need for training on how to do this kind of development, and even just basic Linux kernel development skills is in very high demand. Hopefully through the new Linux Foundation effort of

providing training, this need can be met. Although I still expect myself and other members of the LDP to continue to travel and speak about this topic.

Goal three (work in the open) has succeeded even better than I had expected. At the Linux Kernel Summit last year, the drivers/staging/ tree was created, allowing us a place within the main Linux kernel tree for drivers to be merged that were not yet of the high kernel quality standards. This has allowed many users the ability to use their hardware with Linux much earlier than previously possible, and it has allowed a common place for the community to contribute patches and fixes for these drivers. This has alleviated the need to hunt over the internet for various drivers.

The staging tree has also provided a place for developers wishing to get into Linux kernel development to easily help out and start working on code. Often times it is only the lack of knowing what to work on that keeps many people from participating in kernel development. This portion of the kernel tree provides ample opportunities for people to help out with.

So, what is next for the LDP? As the previous three goals worked so well this past year, I propose just continuing them in the following manner:

- We will work with any company that wishes to have Linux kernel drivers written for them, doing the work, for free, and merging the result into the mainline kernel tree.
- We will work with any company that wishes to have us work to merge their existing Linux kernel drivers into the main kernel tree.
- We will maintain the staging tree, containing "pre-mergeable" code, to provide a centralized point for users to use new hardware sooner, and for developers to be able to join the kernel community.
- We will work on education and training of companies and developers about how the Linux kernel development process works, and help them achieve the technical knowledge to participate fully within it.

I'd first like to thank my employer, Novell, for giving me the opportunity to work on this project full time. Their acceptance and support for the LDP is amazing and has been what has allowed it to survive and produce such great results.

I'd also like to thank all of the developers who have offered to help out with this project. Your volunteering to participate is amazing and shows the strength of the Linux developer community.

I'd also like to thank Tomasz Grzegorzko for maintaining and keeping the linuxdriverproject.org domain and server up and running so well. Also thanks to the OSU Open Source Labs for your domain and bandwidth support of the project.

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Q What do you say about hacks like ndiswrapper, technically as well as legally?

Technically, ndiswrapper is wonderful. The ability to run a driver written for a different operating system, on Linux, is a wonderful engineering achievement. I can't say enough good things about that hack, and I love it. Also, from a legal standpoint, there is nothing wrong with it.

As a long-term solution for users to rely on, that's not a good idea, as remember, the driver was not originally written for, or even tested with, Linux. So it

usually does not work as well as a native Linux driver does and can. 

Interview by Swapnil Bhartiya

A Free Software fund-a-mental-ist and Charles Bukowski fan, Swapnil also writes fiction and tries to find cracks in a proprietary company's 'paper armours'. He is a big movie buff, and prefers listening to music at such loud volumes that he's gone partially deaf when it comes to identifying anything positive about proprietary companies. Oh, and he is also the assistant editor of *EFYTimes.com*.